TM Forum Component

Service Test Management

TMFC055

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Notice

Direct inquiries to the TM Forum office:

100 Enterprise Drive
Suite 301 #1649
Rockaway, NJ 070866, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

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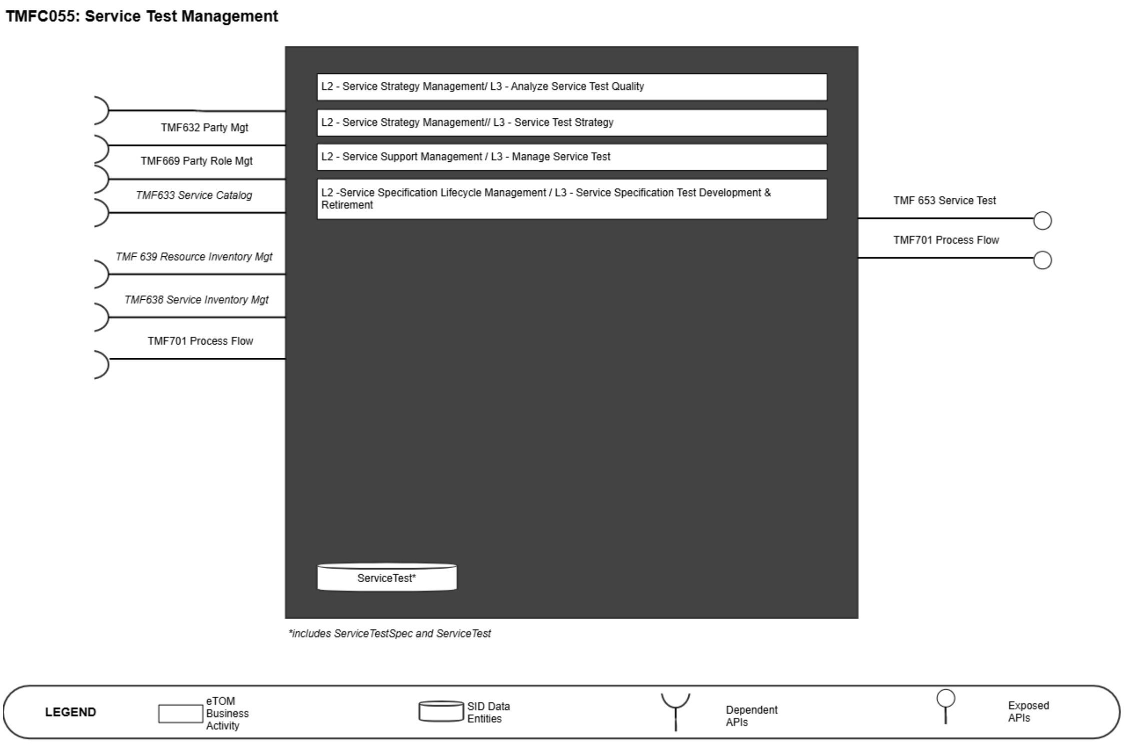
1. Overview

Component Name	ID	Description	ODA Function Block
Service Test Management	TMFC055	**Service Test Management** is responsible for managing the service test specification and executing tests based on this specification to ensure the service operates correctly. Service Test Specification involves capturing potential tests for each Service Specification that can be initiated either automatically or manually by the service user. These tests are derived from those available at the service specification or resource specification level. The scope of the component includes the end-to-end execution and reporting of tests, tailored to the specific context and actor involved.	Production



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

<Note to not be inserted onto ODA Component specifications: If a new ABE is required, but it does not yet exist in SID. Then you should include a textual description of the new ABE, and it should be clearly noted that this ABE does not yet exist. In addition a Jira epic should be raised to request the new ABE is added to SID, and the SID team should be consulted. Finally, a decision is required on the feasibility of the component without this ABE. If the ABE is critical then the component specification should not be published until the ABE issue has been resolved. Alternatively if the ABE is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.4.1	*L2*	Service Strategy Management	Enable the development of a strategic view and a multi-year business plan for the enterprise's services and service directions, and the parties who will supply the required services.
1.4.1.8	*L3*	Service Test Strategy	Service Test Strategy develops the strategies of the enterprise for Service Test. This process is in charge of identifying types of Service Test to be conducted according to different context (i.e. business activities) for types of Services. The different context for Service Test are: - Service Development & Retirement to qualify the capacity to deliver Services before validating a new ServiceSpecification - Service Configuration & Activation to test the Service before closing the ServiceOrderItem - Service Quality Management to test Service Quality - Service Problem Management to test Service functioning - Service Test Management to conduct tests not specific to a customer's product

Identifier	Level	Business Activity Name	Description
1.4.1.9	L3	Analyze Service Test Quality	Analyze Service Test Quality process performs quality analysis of service testing processes, in order to continuously fine-tune and improve them. Service Test Quality Analysis processes provide an offline analysis of the performance of service testing processes to ensure their continuous fitness for purpose, and eventually propose modifications or enhancements to the current service test specifications (including: service testing methods, quota, authorized users, recommendations and course of action if needed). The analysis is done using statistical service test usage data to evaluate service tests performance and required improvements if needed. Following the analysis, typical course of action can be: - No changes needed currently, - Creation of additional service tests, - Improving existing service tests, - Removing service tests that are no longer relevant.
1.4.4	L2	Service Support Management	Manage service infrastructure, ensuring that the appropriate service capacity is available and ready to support the SM&O Fulfillment, Assurance and Billing processes
1.4.4.6	L3	Manage Service Test	Manage Service Test process manages the end-to-end execution of a test or test scenario for services not specific to a customer. Tests can be manual or automated. Service Test Management processes rely on Product and Resource Test Management processes due to dependencies between product tests and resource tests. Service Test Management includes: - Identification of service tests needed according to fulfillment and assurance issues - triggering of service tests in manual or automated mode - Execution of service tests - Verification of test authorization (role and context) and quota management - Identification and prioritization of tests - Setting up the test context and configuration - Triggering appropriate product and resource Tests - Tests can be on-demand or planned according to specific needs - Enrichment with product and resource tests results

Identifier	Level	Business Activity Name	Description
			based on applicable service test rules - Reporting of test results
1.4.3	L2	Service Specification Lifecycle Management	Service Specification Lifecycle Management processes are project oriented in that they develop and deliver new or enhanced service types. These processes include process and procedure implementation, systems changes and customer documentation. They also undertake rollout and testing of the service type, capacity management and costing of the service type. It ensures the ability of the enterprise to deliver service types according to requirements. This process was renamed in 24.0. Old name was- Service Specification Development & Retirement
1.4.3.8	L3	Service Specification Test Development & Retirement	Service Test Development & Retirement is in charge of the Service Test catalogue. A type of Service Test aims at measuring proper functioning and capacities of a Service. Service Test Development & Retirement includes: - Specifying in detail each Service Test according to the different context. It includes specifying: - the roles authorized to use the Test and quotas for each type of role - the method to conduct the Test - the rules that define the strategies for conducting the test (including the test plan) - the thresholds and related actions - the relationships with lower level tests (Resource Test) - the report of test results with rules for enrichment of Resource Tests results according to role asking for it - Specifying test scenarios defining sequence of Tests with rules about context and planning to trigger it. It includes roles allowed for asking test scenario and corresponding quotas.

2.2. SID ABEs

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component specification should not be published until the ABE issue has been resolved. Alternatively if the ABE is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Service Domain.Service Test ABE	ServiceTest
	ServiceTestSpec

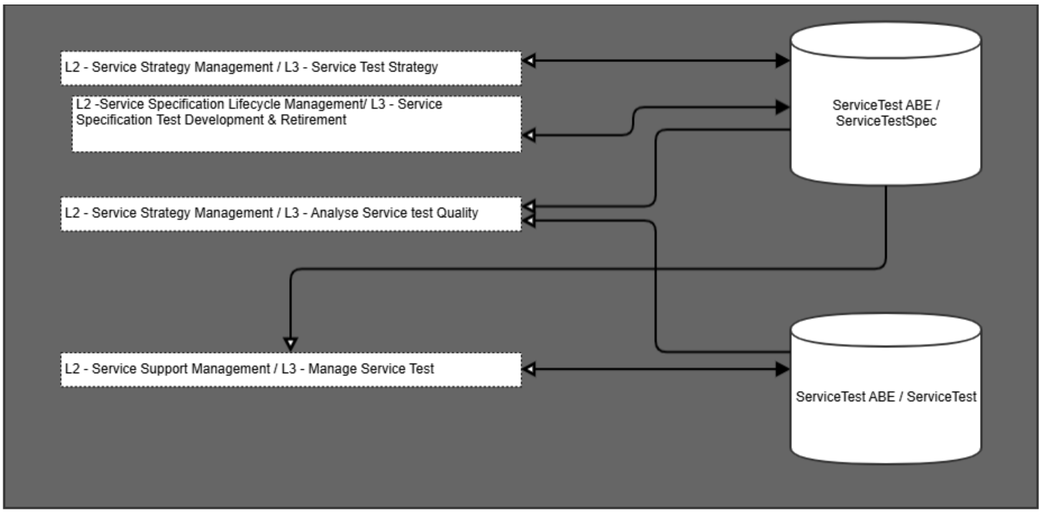
*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

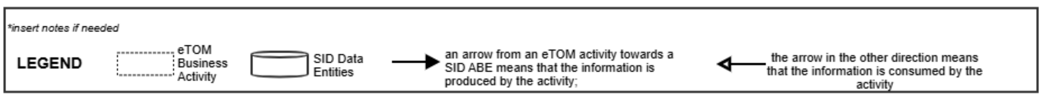
[ISA-1318] Verify role of Test Measure, Test Measure Definition and Test Specification in ServiceTestABE - TM Forum JIRA

Note: Jira ISA-1318 is raised for open point - if Test Measure, Test Measure Definition and Test Specification Role to be included in ServiceTestABE under Service, Resource, Product tabs.

2.3. eTOM L2 - SID ABEs links

eTOM L2 Vs SID ABEs links for this ODA Component.





2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	AF L1 ID	Aggregate Function Level 2	AF L2 ID
625	Service Test Operation Control	Service Test Operation Control function provides the necessary functionality to access, command and control the various service test devices required to perform service testing.	Service Capability Management	4.2.1	Service Test Resource Check	4.2.1.1
1136	Service Test Specification and Scenario Management	Service Test Specification and Scenario Management Function includes: Test scenario management: for example, a home equipment self-test can trigger a test scenario that verifies the functioning of all the services associated, depending on the installed products of the user. Thus, the box could test its internal configuration and functioning and its capacity to connect to internet and VoIP services. Managing service test capabilities: some resources used to implement services have limited capabilities for executing tests, especially for multiple parallel testing. All the constraints linked to	Service Test Specification Development	4.3.3	Service Test Policy Management	4.3.3.1

Function ID	Function Name	Function Description	Aggregate Function Level 1	AF L1 ID	Aggregate Function Level 2	AF L2 ID
		limited resources test capabilities and depending on the execution context must be aggregated and managed at service test level. Managing initiator authorization: some tests are restricted to some user roles (internal or external) so the authorization are defined at the Service Test Strategy and Policy Management.				
566	Service Test Result Analysis Policy Configuration	Service Test Result Analysis Policy Configuration functions provide the necessary functionality to manage how the test results should be interpreted. Strategies can range from simple (e.g., - check the status of a router) to complex (e.g., - perform an end-to-end test on a circuit and sectionalize the problem).	Service Test Specification Development	4.3.3	Service Test Policy Management	4.3.3.1
573	Service Testing Rules Management	Service Testing Rules Management provides management of service testing rules.	Service Test Specification Development	4.3.3	Service Test Policy Management	*4.3.3.1*
581	Automated Service Test Invocation	Automated Service Test Invocation provides automated invocation of Service Test Services for test and retrieval of	Service Test Management	4.4.2	Service Test Conducting	4.4.2.2

Function ID	Function Name	Function Description	Aggregate Function Level 1	AF L1 ID	Aggregate Function Level 2	AF L2 ID
		results from the resource testing capabilities.				
1138	Service Test Reporting Rules Management	Service Test Reporting Rules Management Function includes: Test rules defining the enrichment of test results. - For example: enrichment of internet bandwidth measure based on installed product Test rules defining the restitution of the result depending on test execution context - For example: a test result of ok/not ok can be enough for a user whereas customer care requires more details in the result of the test.	Service Test Specification Development	4.3.3	Service Test Policy Management	4.3.3.1
582	Manual Service Test Invocation	Manual Service Test Invocation provides a user interface (GUI) function that provides the means to access the testing capabilities.	Service Test Management	4.4.2	Service Test Conducting	4.4.2.2
627	Service Test Strategy Management	Service Test Strategy Management functions provide the necessary functionality to manage the local rules that define the strategies for conducting a test as well as how the test	Service Strategy Management	4.1.1	Service Strategy Design and Planning	4.1.1.3

Function ID	Function Name	Function Description	Aggregate Function Level 1	AF L1 ID	Aggregate Function Level 2	AF L2 ID
		results should be interpreted.				
1137	Service Test Rules Management	Service Test Rules Management Function includes: Test rules defining the strategies for setting up test configuration Test rules defining how to carry out the service test. - For example: to choose the next test in a scenario depending on the results of the previous tests.	Service Test Specification Development	4.3.3	Service Test Policy Management	4.3.3.1
575	Service Test Results Reporting	Service Test Results Reporting provides reporting of service end to end test results back to the client.	Service Test Management	4.4.2	Service Test Reporting	4.4.2.4
626	Service Testing Management	Service Testing Management functions provides the necessary functionality to manage the end-to-end lifecycle of a test of a service. Including the management of scheduling, retrieval of appropriate inventory data, setting up the test configuration, acquisition and management of test resources, test results interpretation, reporting of test results and close	Service Test Management	4.4.2	Service Test Repository Management	4.4.2.3

Function ID	Function Name	Function Description	Aggregate Function Level 1	AF L1 ID	Aggregate Function Level 2	AF L2 ID
		down of the test of the service.				
577	Service End to End Test Scheduling	Service End to End Test Scheduling provides scheduling of end to end testing of services.	Service Test Specification Development	4.3.3	Service Test Policy Management	4.3.3.1
579	Service End to End Testing	Service End to End Testing provides test execution of the end to end service testing.	Service Test Management	4.4.2	Service Test Conducting	4.4.2.2

3. TM Forum Open APIs & Events

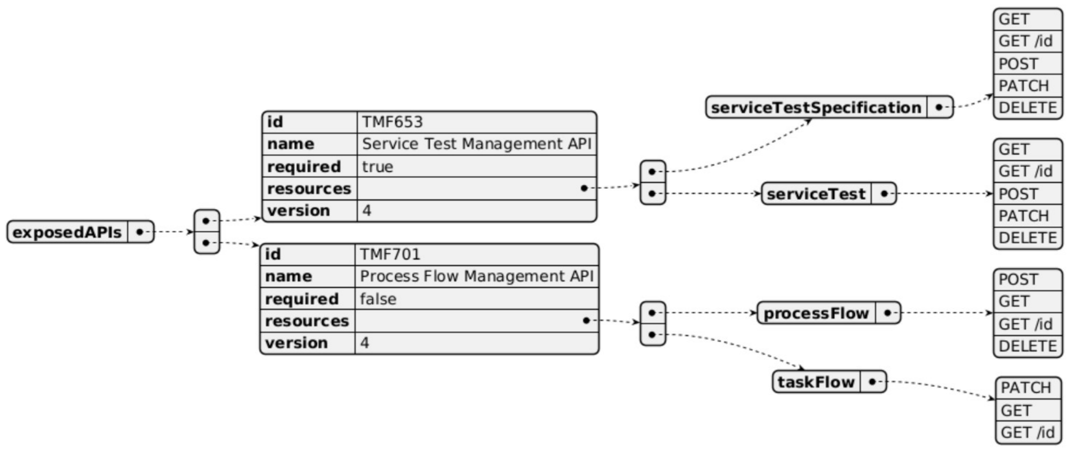
The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

<Note note to be inserted into ODA Component specifications: If a new Open API is required, but it does not yet exist. Then you should include a textual description of the new Open API, and it should be clearly noted that this Open API does not yet exist. In addition, a Jira epic should be raised to request the new Open API is added, and the Open API team should be consulted. Finally, a decision is required on the feasibility of the component without this Open API. If the Open API is critical then the component specification should not be published until the Open API issue has been resolved. Alternatively if the Open API is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

3.1. Exposed APIs

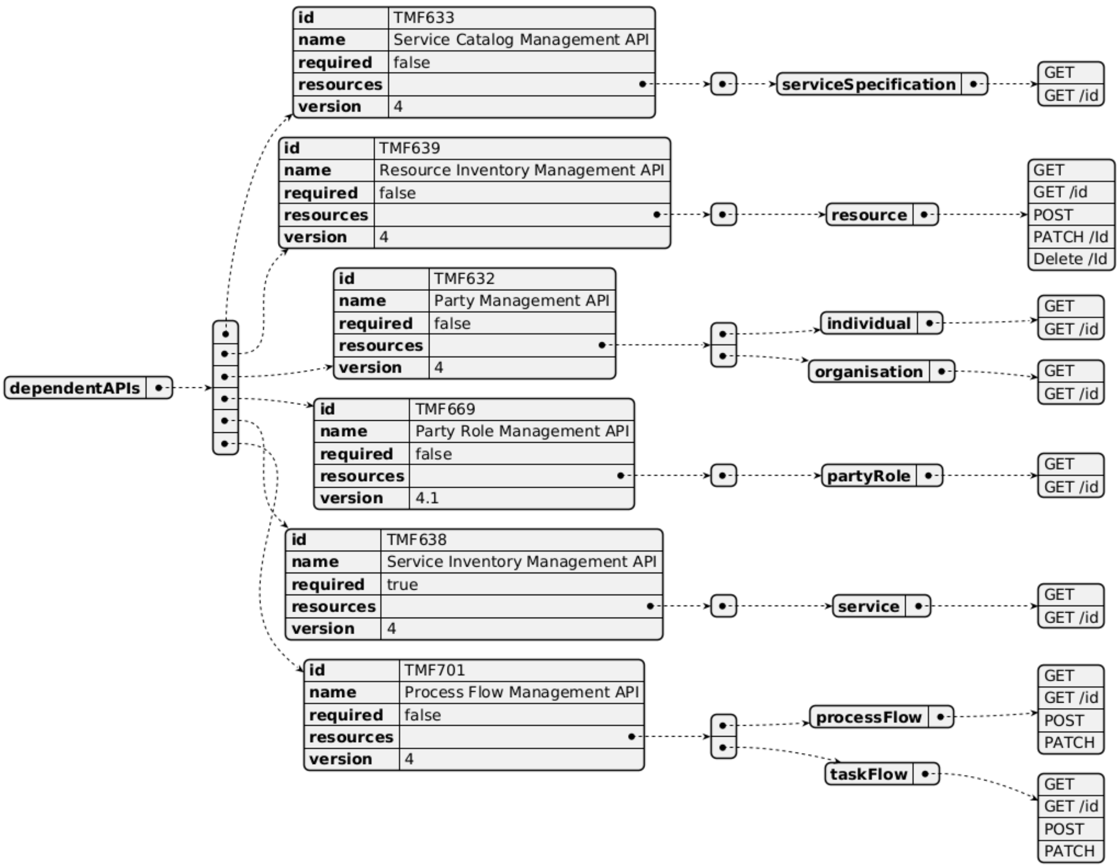
Following diagram illustrates API/Resource/Operation:



API ID	API Name	API Version	Mandatory / Optional	Operations
TMF 653	Service Test Management API	*V4*	mandatory	ServiceTestSpecification : GET, GET /id, POST, PATCH, DELETE ServiceTest : GET, GET /id, POST, PATCH, DELETE
TMF701	Process Flow	V4	optional	processFlow: POST, GET, GET /id, DELETE taskflow: PATCH, GET, GET /id

3.2. Dependent APIs

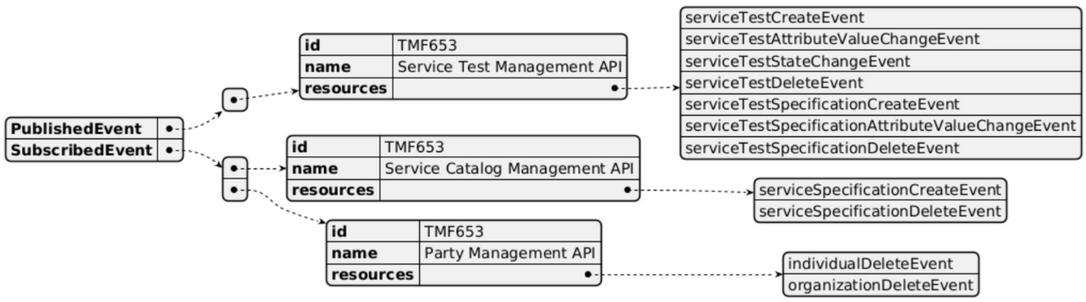
Following diagram illustrates API/Resource/Operation potentially used by the product catalog component:



API ID	API Name	API Version	Mandatory / Optional	Operations	Rationale
TMF632	Party Management	*V4*	Optional	individual: GET, GET /id organisation: GET, GET /id	
TMF669	Party Role Management	*V4*	Optional	partyRole: GET, GET /id	
TMF633	Service Catalog	*V4*	*Optional*	*serviceSpecification : GET, GET /id*	*a ServiceTestSpecification must be associated to an existing SpecificationSpecification*
TMF638	Service Inventory Management	*V4*	*Mandatory*	*service: GET, GET /id*	*ServiceTest must correspond to an existing Service*
TMF 639	Resource Inventory	*V4*	*Optional*	resource: GET, GET /id, POST, PATCH, DELETE/id	
TMF701	Process Flow	*V4*	*Optional*	processFlow: GET, GET /id, POST, PATCH, taskFlow: GET, GET /id, POST, PATCH,	

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	*24.5*
eTOM	*24.5*
Functional Framework	*24.5*

5.2. Jira References

Note: Same needs as at Service level for non-tangible products + add of Resource Inventory APIs to be consistent.

5.2.1. SID

[ISA-1251] Improve the description of Product/Service/Resource Test ABEs - TM Forum JIRA

The links between ServiceTest and ResourceTest need to be described, at specification and instance levels. For example

- a ServiceTestSpecification related to a Service should correspond to an existing ResourceTestSpecification
- a ServiceTest related to a Service should trigger a ResourceTest on the corresponding Resource.

5.2.2. APIs

As

- a ServiceSpecification related to a Service should correspond to an existing ResourceTestSpecification;
- a ServiceTest related to a Service should trigger a ResourceTest on the corresponding Resource.

[AP-6690] TMF769 - add reference to ServiceTest and ResourceTest - TM Forum JIRA: TMF769 Product Test API is only available in preview - but contains no links to Service Test or Resource Test, where the real tests are done.

[AP-6691] New API : Resource Test API - TM Forum JIRA: No Resource Test Management API identified yet, while we should be able to check that:

5.3. Further resources

1. IG1242: please refer to IG1242 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	18-Jul-2025	Rosie Wilson	Final Administrative Updates

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	18-Jul-2025	Rosie Wilson	Initial release

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

Member	Company	Role*
Ritu Arora	British Telecom	Author
Milind Bhagwat	British Telecom	Reviewer
Marcio Sousa	Altice Labs	Additional Input
Sylvie Demarest	Orange	Reviewer
Anastasios Sarantis	City Fibre	Reviewer

**Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer*